

Report
Of
SmartSplit: A Full-Stack Expense Management and
Payment Settlement Application

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

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Batch: 2026-27

CANDIDATE'S DECLARATION

I, **KASHISH**, do hereby declare that the internship report titled “*SmartSplit: A Full-Stack Expense Management and Payment Settlement Application*” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: KASHISH

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ACKNOWLEDGEMENT

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I would also like to thank the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their guidance and encouragement throughout this internship.

I express my gratitude to my parents for their blessings and constant support.

Signature: _____

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Roll No.: CS2341300

INTERNSHIP COMPLETION CERTIFICATE

The Internship Completion Certificate issued by BitePay, confirming successful completion of the two-month internship as an AI and Data Analytics Intern, is provided below.



Fig. i — BitePay Internship Completion Certificate

The original Summer Internship Offer Letter issued by BitePay at the start of the internship is also included below for reference.



Fig. ii — BitePay Summer Internship Offer Letter

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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the work carried out during a two-month Summer Internship undertaken as part of the B.Tech Computer Science and Engineering programme at IILM University, Greater Noida. The internship was completed at BitePay, a fintech company, from 7 June 2026 to 7 August 2026, in the domain of Artificial Intelligence, Data Analytics, and Software Development.

During the internship, the primary project undertaken was SmartSplit, a full-stack expense-splitting and payment-settlement web application. The project was designed to simplify the process of managing shared expenses among a group of people by allowing users to create shared expenses, split amounts either equally or by custom shares, track balances on a dashboard, and settle outstanding dues through an integrated online payment gateway. This report documents the organization profile, problem statement, objectives, scope, technologies used, system architecture, development methodology, and outcomes of the project.

4.2 Organization Profile

BitePay (“Tech that Serves”) is a fintech company that builds digital payment and financial-utility products, working across Artificial Intelligence, Data Analytics, and software engineering to power its payment-related software. The organization is registered under MSME (2025) and focuses on practical, real-world product building, with interns contributing directly to live features and ongoing development work.

As an intern in the AI, Data Analytics & Software Development domain, the author contributed to building a fintech-style expense and payment-settlement product from the ground up, working under the guidance of the BitePay team.

4.3 Problem Statement

Managing shared expenses among groups of people — such as roommates, friends on a trip, or colleagues sharing a bill — is a common but often disorganized task. Expenses are typically tracked informally through messaging apps, spreadsheets, or memory, which frequently leads to confusion about who owes whom, disputes over amounts, and delayed or forgotten settlements. There is a need for a simple, reliable, and digitally verifiable system that allows a group to record shared expenses, calculate individual

shares accurately, and settle outstanding balances securely online, without depending on manual tracking or trust-based verbal agreements.

4.4 Project Objectives

1. To design and develop a full-stack web application for managing shared expenses among groups of users.
2. To implement secure user registration and login functionality using JWT-based authentication.
3. To enable the creation of shared expenses with both equal and custom splitting among participants.
4. To provide users with a dashboard to track their expenses and outstanding balances.
5. To integrate a real-world online payment gateway (Razorpay) for settling outstanding shares.
6. To implement backend verification of payments so that the application does not rely solely on frontend confirmation of a successful transaction.

4.5 Scope of the Project

The scope of the SmartSplit project, in its current version, includes the following:

- User registration and secure login using JWT-based authentication and bcrypt password hashing.
- Creation of shared expenses and division of amounts among participants, either equally or by custom shares.
- A dashboard for users to view their expenses and outstanding balances.
- Initiation and settlement of payments for outstanding shares through Razorpay Checkout.
- Backend verification of payment details returned from the payment gateway, rather than trusting frontend confirmation alone.

The following are considered outside the scope of the current version of the project and may be taken up as future enhancements: multi-currency support, recurring or subscription-based expenses, a dedicated native mobile application, push

notifications/reminders for pending dues, and advanced analytics or spending-report generation beyond the existing dashboard.

4.6 Technologies and Tools Used

The following technologies and tools were used in the development of the SmartSplit application:

Category	Technologies / Tools
Frontend	React.js, JavaScript, HTML, CSS
Backend	Node.js, Express.js
Database	MongoDB with Mongoose ODM
Authentication	JWT (JSON Web Tokens)
Security	Password hashing using bcrypt
Payments	Razorpay Payment Gateway and Razorpay Checkout
API	RESTful APIs built using Express
Development Tools	VS Code, npm, Nodemon, Git/GitHub
Environment Management	dotenv
Other	CORS, browser Local Storage (for maintaining login sessions)

Table 4.1 — Technologies and Tools Used in SmartSplit

4.7 System Architecture

The SmartSplit application follows a client-server architecture. The React.js frontend communicates with a Node.js/Express.js backend through RESTful APIs. The backend handles authentication using JWT and bcrypt, interacts with a MongoDB database (via Mongoose) for storing user, expense, and transaction data, and communicates with the Razorpay payment gateway to create payment orders and verify completed payments. Figure 4.1 illustrates this architecture.

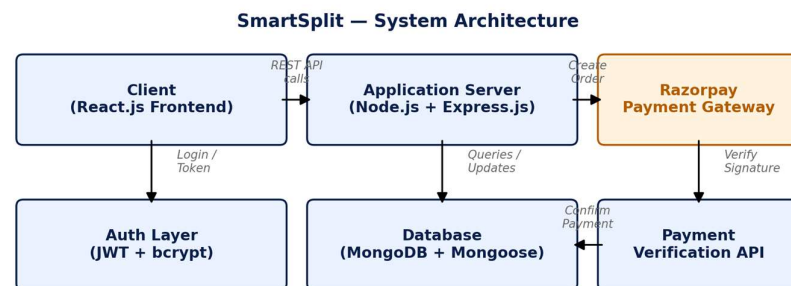


Figure 4.1 — System Architecture of SmartSplit

4.8 Methodology

The project was developed following an incremental, module-by-module methodology, integrating the frontend, backend, database, and payment workflow into a single working application rather than building components in isolation. Figure 4.2 illustrates the overall development workflow followed during the internship.

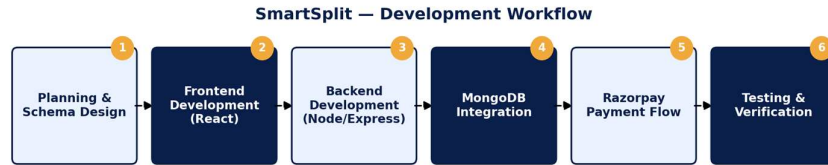


Figure 4.2 — Development Workflow of SmartSplit

During development, one of the major challenges encountered was integrating MongoDB Atlas with the Node.js backend, particularly DNS/SRV resolution and TLS connection issues while connecting to the Atlas cluster. Debugging this issue helped in understanding how database connection strings, DNS resolution, network access/IP whitelisting, and Node.js configuration affect a backend application. Another significant challenge was integrating the Razorpay payment flow, particularly around creating orders, opening Razorpay Checkout from the React frontend, and securely verifying payments on the backend rather than assuming success on the client side.

4.9 Expected Outcomes

Upon completion of the internship, the following outcomes were achieved:

- A working full-stack SmartSplit prototype comprising a React dashboard and a Node.js/Express backend connected to MongoDB.
- Users can register, log in, manage shared expenses, split expenses between participants, and view outstanding amounts.
- Users can initiate payments for their outstanding shares through Razorpay, with payment details verified on the backend before being marked as successful.
- Practical, hands-on experience in building and integrating a complete fintech-style application, combining authentication, database operations, expense management, and online payments.

- A stronger understanding of full-stack development as an integrated process — connecting the frontend, backend, database, and third-party services — rather than working on isolated components.

4.10 Certificates of Completion and Communication Proof

The Internship Completion Certificate issued by BitePay on 17 August 2026, confirming successful completion of the two-month internship as an AI and Data Analytics Intern, along with the original Summer Internship Offer Letter, has been attached in the front matter of this report (see the Internship Completion Certificate section).

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